

Wigner 3-j and 6-j symbols

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Recoupling of angular momenta

Clebsch-Gordan coefficient:

$$\langle j_1, m_1, j_2, m_2 | j_1, j_2, j, m \rangle = (-1)^{j_1-j_2+m} \sqrt{2j+1} \begin{pmatrix} j_1 & j_2 & j \\ m_1 & m_2 & -m \end{pmatrix}$$

Recoupling coefficient for three angular momenta:

$$\langle (j_1 j_2) j_{12}, j_3, j | j_1, (j_2 j_3) j_{23}, j \rangle = \sqrt{(2j_{12}+1)(2j_{23}+1)} (-1)^{j_1+j_2+j_3+j} \begin{Bmatrix} j_3 & j & j_{12} \\ j_1 & j_2 & j_{23} \end{Bmatrix}$$

Judd 1963: (3-4)

Wigner 3-j symbol: Definition

$$\begin{aligned}
 \begin{pmatrix} j_1 & j_2 & j_3 \\ m_1 & m_2 & m_3 \end{pmatrix} &= \delta(m_1 + m_2 + m_3) (-1)^{j_1 - j_2 - m_3} \\
 &\times \sqrt{\frac{(j_1 + j_2 - j_3)!(j_1 - j_2 + j_3)!(-j_1 + j_2 + j_3)!(j_1 - m_1)!}{(j_1 + m_1)!(j_2 - m_2)!(j_2 + m_2)!(j_3 - m_3)!(j_3 + m_3)!}} \\
 &\times \sum_k \frac{(-1)^k}{\binom{k!(j_1 + j_2 - j_3 - k)!(j_1 - m_1 - k)!(j_2 + m_2 - k)!}{(j_3 - j_2 + m_1 + k)!(j_3 - j_1 - m_2 + k)!}}
 \end{aligned}$$

(Fritzsche 1997)

Wigner 3-j symbol: Symmetry properties

Even permutations of columns do not change the value:

$$\begin{pmatrix} j_1 & j_2 & j_3 \\ m_1 & m_2 & m_3 \end{pmatrix} = \begin{pmatrix} j_2 & j_3 & j_1 \\ m_2 & m_3 & m_1 \end{pmatrix} = \begin{pmatrix} j_3 & j_1 & j_2 \\ m_3 & m_1 & m_2 \end{pmatrix}$$

Odd permutations of columns introduce a phase factor:

$$\begin{pmatrix} j_1 & j_2 & j_3 \\ m_1 & m_2 & m_3 \end{pmatrix} = (-1)^{j_1+j_2+j_3} \begin{pmatrix} j_2 & j_1 & j_3 \\ m_2 & m_1 & m_3 \end{pmatrix}$$

Multiplying the bottom line by -1 introduces the same phase factor:

$$\begin{pmatrix} j_1 & j_2 & j_3 \\ m_1 & m_2 & m_3 \end{pmatrix} = (-1)^{j_1+j_2+j_3} \begin{pmatrix} j_1 & j_2 & j_3 \\ -m_1 & -m_2 & -m_3 \end{pmatrix}$$

Wigner 3-j symbol: Special values I

If $J = j_1 + j_2 + j_3$ is even, then

$$\begin{aligned} \begin{pmatrix} j_1 & j_2 & j_3 \\ 0 & 0 & 0 \end{pmatrix} &= (-1)^{\frac{1}{2}J} \sqrt{\frac{(J - 2j_1)!(J - 2j_2)!(J - 2j_3)!}{(J + 1)!}} \\ &\quad \times \frac{(\frac{1}{2}J)!}{(\frac{1}{2}J - j_1)!(\frac{1}{2}J - j_2)!(\frac{1}{2}J - j_3)!} \\ &\quad \text{(Judd 1963: (4-3))} \end{aligned}$$

If J is odd, the value of the above 3-j symbol is zero.

$$\begin{pmatrix} l & k & l \\ 0 & 0 & 0 \end{pmatrix} = (-1)^{l+\frac{k}{2}} \sqrt{\frac{(2l - k)!}{(2l + k + 1)!}} \frac{k!(l + \frac{k}{2})!}{(\frac{k}{2}!)^2(l - \frac{k}{2})!}$$

$$\begin{pmatrix} j & j' & 0 \\ m & -m & 0 \end{pmatrix} = \begin{pmatrix} j' & 0 & j \\ -m & 0 & m \end{pmatrix} = \delta(j, j') \frac{(-1)^{j-m}}{\sqrt{2j+1}}$$

Wigner 3-j symbol: Special values II

$$\begin{pmatrix} 0 & k & k \\ 0 & 0 & 0 \end{pmatrix} = \frac{(-1)^k}{\sqrt{2k+1}}$$

$$\begin{pmatrix} 1 & k & k-1 \\ 0 & 0 & 0 \end{pmatrix} = (-1)^k \sqrt{\frac{k}{(2k+1)(2k-1)}}$$

$$\begin{pmatrix} 1 & k & k \\ 0 & 0 & 0 \end{pmatrix} = 0$$

$$\begin{pmatrix} 1 & k & k+1 \\ 0 & 0 & 0 \end{pmatrix} = (-1)^{k+1} \sqrt{\frac{k+1}{(2k+3)(2k+1)}}$$

Wigner 3-j symbol: Reduction of two 3-j symbols

$$\sum_{j_3, m_3} (2j_3 + 1) \begin{pmatrix} j_1 & j_2 & j_3 \\ m_1 & m_2 & m_3 \end{pmatrix} \begin{pmatrix} j_1 & j_2 & j_3 \\ m'_1 & m'_2 & m_3 \end{pmatrix} = \delta(m_1, m'_1) \delta(m_2, m'_2)$$

Judd 1963: (1-23)

$$\sum_{m_1, m_2} \begin{pmatrix} j_1 & j_2 & j_3 \\ m_1 & m_2 & m_3 \end{pmatrix} \begin{pmatrix} j_1 & j_2 & j'_3 \\ m_1 & m_2 & m'_3 \end{pmatrix} = \frac{\delta(j_3, j'_3) \delta(m_3, m'_3)}{(2j_3 + 1)}$$

Judd 1963: (1-24)

Wigner 6-j symbol: Definition

$$\left\{ \begin{matrix} j_1 & j_2 & j_3 \\ l_1 & l_2 & l_3 \end{matrix} \right\} = \Delta(j_1 j_2 j_3) \Delta(j_1 l_2 l_3) \Delta(l_1 j_2 l_3) \Delta(l_1 l_2 j_3) \\ \times \sum_z \frac{(-1)^z (z+1)!}{\left[\begin{aligned} &(z - j_1 - j_2 - j_3)! (z - j_1 - l_2 - l_3)! (z - l_1 - j_2 - l_3)! \\ &\times (z - l_1 - l_2 - j_3)! (j_1 + j_2 + l_1 + l_2 - z)! \\ &\times (j_2 + j_3 + l_2 + l_3 - z)! (j_3 + j_1 + l_3 + l_1 - z)! \end{aligned} \right]}$$

$$\Delta(abc) = \sqrt{\frac{(a+b-c)!(a-b+c)!(-a+b+c)!}{(a+b+c+1)!}}$$

Judd 1963: (3-7)

Wigner 6-j symbol: Symmetry properties

Permutations of columns do not change the value:

$$\begin{Bmatrix} j_1 & j_2 & j_3 \\ l_1 & l_2 & l_3 \end{Bmatrix} = \begin{Bmatrix} j_2 & j_1 & j_3 \\ l_2 & l_1 & l_3 \end{Bmatrix}$$

Judd 1963: (3-8)

Exchange of upper and lower argument in each of any pair of columns does not change the value:

$$\begin{Bmatrix} j_1 & j_2 & j_3 \\ l_1 & l_2 & l_3 \end{Bmatrix} = \begin{Bmatrix} l_1 & l_2 & j_3 \\ j_1 & j_2 & l_3 \end{Bmatrix}$$

Judd 1963: (3-9)

Wigner 6-j symbol: Special values

$$\left\{ \begin{matrix} 0 & k & k \\ l & l & l \end{matrix} \right\} = \frac{(-1)^{2l+k}}{\sqrt{(2k+1)(2l+1)}}$$

$$\left\{ \begin{matrix} 1 & k & k-1 \\ l & l & l \end{matrix} \right\} = \frac{(-1)^{2l+k}}{2} \sqrt{\frac{(2l+k+1)(2l-k+1)k}{l(l+1)(2l+1)(2k-1)(2k+1)}}$$

$$\left\{ \begin{matrix} 1 & k & k \\ l & l & l \end{matrix} \right\} = \frac{(-1)^{2l+k+1}}{2} \sqrt{\frac{k(k+1)}{l(l+1)(2l+1)(2k+1)}}$$

$$\left\{ \begin{matrix} 1 & k & k+1 \\ l & l & l \end{matrix} \right\} = \frac{(-1)^{2l+k+1}}{2} \sqrt{\frac{(2l+k+2)(2l-k)(k+1)}{l(l+1)(2l+1)(2k+1)(2k+3)}}$$

Judd 1963: (3-10)

Conversion equations

$$\left\{ \begin{matrix} j_1 & j_2 & j_3 \\ l_1 & l_2 & l_3 \end{matrix} \right\} \left(\begin{matrix} j_1 & j_2 & j_3 \\ m_1 & m_2 & m_3 \end{matrix} \right) = \sum_{\mu_1, \mu_2, \mu_3} (-1)^{l_1+l_2+l_3+\mu_1+\mu_2+\mu_3} \\ \times \left(\begin{matrix} j_1 & l_2 & l_3 \\ m_1 & \mu_2 & -\mu_3 \end{matrix} \right) \left(\begin{matrix} l_1 & j_2 & l_3 \\ -\mu_1 & m_2 & \mu_3 \end{matrix} \right) \left(\begin{matrix} l_1 & l_2 & j_3 \\ \mu_1 & -\mu_2 & m_3 \end{matrix} \right)$$

Judd 1963: (3-6)

$$\sum_{m_2} (-1)^{m_2} \left(\begin{matrix} l & k_1 & l \\ -m_3 & q_1 & m_2 \end{matrix} \right) \left(\begin{matrix} l & k_2 & l \\ -m_2 & q_2 & m_1 \end{matrix} \right) = \\ \sum_{k_3, q_3} (-1)^{l+k_1+k_2+k_3-q_3} (2k_3 + 1) \\ \times \left\{ \begin{matrix} k_1 & k_2 & k_3 \\ l & l & l \end{matrix} \right\} \left(\begin{matrix} l & k_3 & l \\ -m_3 & q_3 & m_1 \end{matrix} \right) \left(\begin{matrix} k_1 & k_2 & k_3 \\ q_1 & q_2 & -q_3 \end{matrix} \right)$$

Judd 1963: (5-14)